Lesson 3 · Good data, fair AI · for your pupils

Whose examples?

40 minutes | Ages 10–16 · adapt for other ages below | No electricity needed | No devices needed | Photocopy-friendly

Big question

Why does AI sometimes work better for some people than others?

Pupils will be able to...

- Explain that a machine that learns from examples can make more mistakes when some examples are missing.
- Spot who or what is missing from a set of examples.
- Suggest a fair way to fill a data gap, such as adding examples from the people left out.

You need

- Board and chalk (or paper on the wall)
- Scrap paper and pencils
- Two or three local fruits, or drawings of them
- Worksheet: Spot the data gap (one per pair, optional)

Before class

Draw a big box on the board labelled “Our examples”. Bring two or three local fruits pupils may not think to draw, such as guava, dates, prickly pear, jackfruit or a baobab pod.

Time	What you do	What you can say	Pupils
Warm-up 0–5 min	Ask: has a phone or app ever got someone’s name or words wrong? Take 3–4 stories. Write “Why?” on the board.	*“Today we find out why some machines work well for some people and badly for others.”*	Share stories from home; guess why it happens.
Draw examples 5–13 min	Each pupil draws one fruit on scrap paper and names it. Put all the drawings in the “Our examples” box. Count each kind together.	*“These drawings are our examples. Our pretend recogniser can only learn from what is in this box.”*	Draw one fruit, add it to the box, and help count each kind.
Test it 13–23 min	The class is the recogniser and may only answer with a fruit in the box. Show a common fruit, then a local fruit nobody drew. Pupils vote for the closest drawing.	*“The recogniser can only match drawings it has. Which is closest? Is that the right answer?”*	Vote, then notice that the recogniser gave a wrong answer.
Pair talk 23–33 min	Pairs discuss: what was missing, and how could we fix it? Then ask: if voice typing learned mostly from some accents, whose voices might it get wrong?	*“Nobody meant to leave that fruit out. So how could we make our examples fairer?”*	Discuss in pairs; share one gap and one fix with the class.
Wrap-up 33–40 min	Add drawings of the missing fruit and test again. Write on the board: “It’s not just how many examples — it’s whose.” Then do the exit ticket.	*“Before we trust a machine, we ask: whose examples did it learn from — and who is missing?”*	Complete the exit ticket.

Check understanding
Listen for “it never saw that fruit” or “those examples were missing”. If pupils say “the machine is silly” or “it’s unfair on purpose”, ask: “What was in its examples? What was missing?”

Exit ticket
One thing missing from our examples, and one way to fix it.

Common mix-ups

Pupils may think…	Help them see…
The machine got it wrong because it’s silly.	It can only match its examples. Point to the box: the fruit it got wrong was never there.
Machines are always fair.	A machine can be wrong more often for some people when its examples leave them out — even if nobody meant it.
Just add more drawings of anything.	Ask: more of the same fruit, or the fruit that was missing? The mix matters, not just the number.

Adapt it

Younger pupils
Ages 6–9: use three real fruits. Pupils draw only two kinds; then show the third and ask why the “machine” can’t name it.

Older pupils and university
Share the 2020 US speech study: about 35 wrong words in 100 for Black speakers vs 19 for white speakers. Pupils plan a fair test for voice typing: who should take part?

Large classes
Groups of six make one example sheet each. Test every sheet with the same new fruit and compare which groups had a gap.

No board or no space
Lay the drawings on a cloth or on the ground outside as the examples pile. Pupils stand beside the drawing they vote for.

Short on time? Do the drawing, the test with one missing fruit, and two minutes of pair talk on voice typing (20 minutes).

What you need to know

An AI system is only as good as its examples. When some people are missing or rare in them, it often makes more mistakes for them — and we can fix those gaps.

Bias (in AI)
When an AI system works worse, or less fairly, for some groups of people than for others. It often starts with gaps in the examples it learned from.

Data gap
When some people, places, accents or languages are missing — or very rare — in the examples an AI system learned from.

Varied (representative) examples
Examples that match the people, places and things a tool will meet in real use — every group is there, with enough examples of each.

My notes — how did it go?